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CS 654 MACHINE LEARNING AND DATA MINING

Jupyter and Pandas Review

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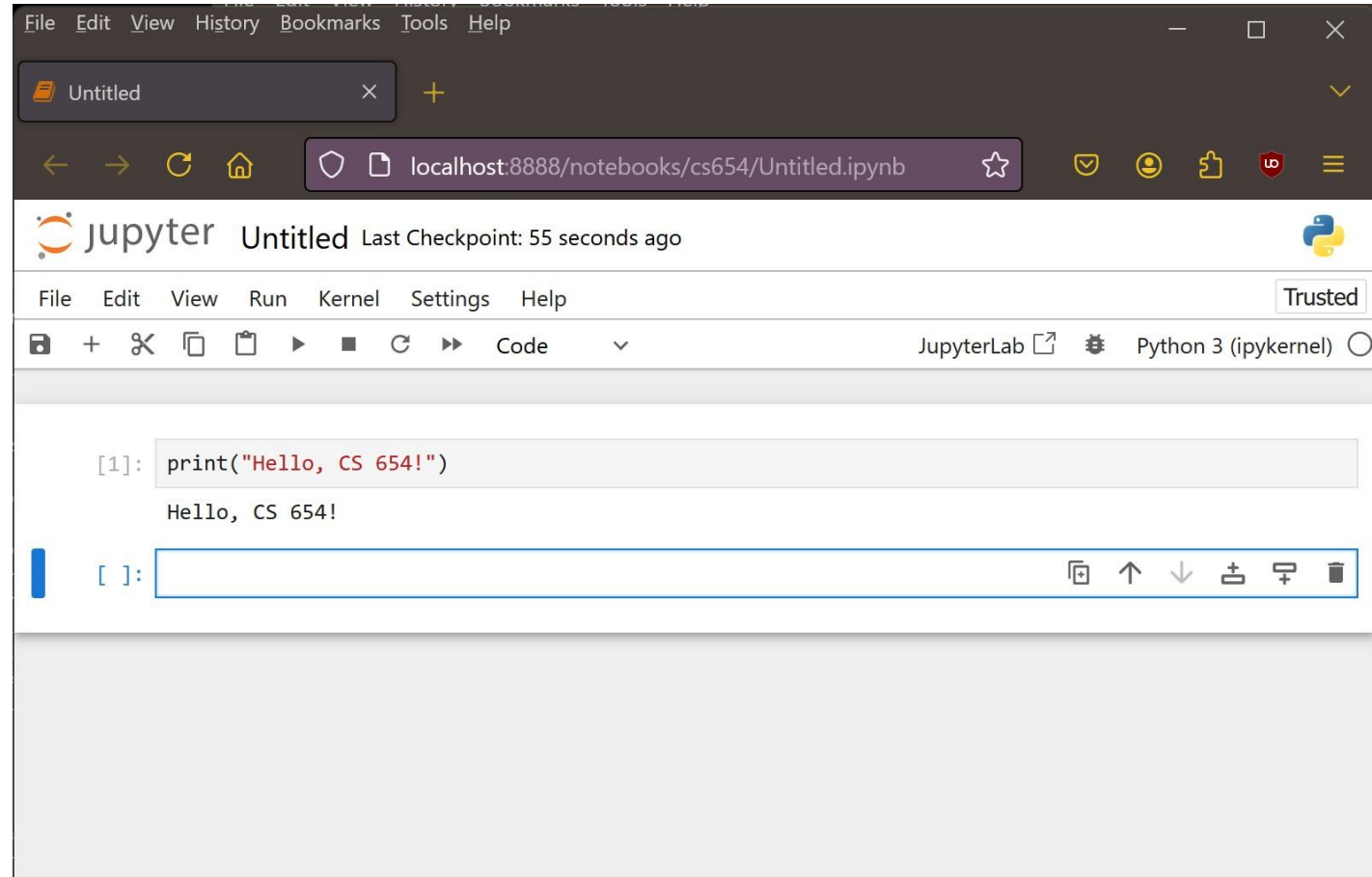
DEVELOPMENT PLATFORM

- Using Anaconda (www.anaconda.com) is encouraged.
- Or install Python (www.python.org) + Jupyter (jupyter.org).
- Our samples will be demonstrated under Jupyter (in a web browser).



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JUPYTER ON A BROWSER



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PANDAS

- Pandas is a data analysis and manipulation tool written in Python.
- Pandas is highly welcome among data scientists.
- <https://pandas.pydata.org>

TWO DATA STRUCTURES IN PANDAS

- A **Series** is one-dimensional array-like object containing an array of data and an associated array of data labels(index).
- A **DataFrame** represents a tabular, spreadsheet-like data structure.

SERIES

- Series is an ordered dictionary.
 - ▣ `>>> from pandas import Series, DataFrame`
- `>>> obj2 = Series([4, 7, -5, 3], index=['d', 'b', 'a', 'c'])`
- `>>> obj2.index`
- `>>> obj2.values`
- `>>> obj2['a']`
- `>>> obj2[2]`



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SERIES CONTD

- `>>> sdata = {'Ohio':35000, 'Texas':71000, 'Oregon':16000, 'Utah':5000}`
- `>>> obj3 = Series(sdata)`
- `>>> obj3 .head(n) #show the first n objects`



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DATAFRAME CONTD

- A DataFrame represents a tabular, spreadsheet-like data structure.
- Often used with data loaded from csv files.

```
>>> from pandas import Series, DataFrame
```

```
>>> data = {'state':['Ohio', 'Ohio', 'Ohio', 'Nevada', 'Nevada'],  
            'year':[2000, 2001, 2002, 2001, 2002], 'pop':[1.5, 1.7, 3.6,  
            2.4, 2.9]}
```

```
>>> frame = DataFrame(data)
```

```
>>> frame
```


DATAFRAME CONTD

```
>>> DataFrame(data, columns=['year', 'state', 'pop'])
```

```
>>> frame2 = DataFrame(data, columns=['year', 'state',  
'pop', 'debt'], index = ['one', 'two', 'three', 'four', 'five'])
```

```
>>> frame2
```



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DATAFRAME CONTD

```
>>> frame2
```

```
>>> frame2.columns
```

```
>>> frame2.index
```

```
>>> frame2.values
```

```
>>> .head(10) #show the first n objects
```

PANDAS DATA ACCESS

- To access an element in a Series:
 - via integer index – `s[i]`
 - via label index – `s['label']`
- To access a row in a dataframe:
 - `df.loc[ [row name]]`(a dataframe), or `df.loc[row name]`(a series)
 - `df.iloc[[row no]]`(a dataframe), or `df.iloc[row no]` (a series)
- To access a column in a dataframe:
 - `df[column name]` (a series)
 - `df[[column name]]` (a dataframe)



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PANDAS DATA ACCESS CONTD

- To access a cell in a dataframe
 - `df.loc[row_name, column_name]`
 - `df.iloc[row_integer_index, column_integer_index]`
- To access a segment in a dataframe
 - `df.loc[:, :, ]` or `df.iloc[:, :, ]`



MOVING DATA BETWEEN DATA STRUCTURES

- Quite often, we need to move data among between data structures. For example:
 - from a series to a list
 - `ser.to_list()`
 - from a dictionary to a dataframe
 - `df.loc[].to_dict()`
- See sample code for demonstration.



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LOAD DATA INTO DATAFRAME

- Use `pandas.read_csv('filename.csv')` to load data stored in a csv file as a dataframe.
- Use `pandas.read_json('filename.json')` to load data stored in a json file as a dataframe.



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